

Introduction to Amazon Simple Storage Service (S3): Hands-on Lab

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Overview

This lab introduces the basic functionality of Amazon S3 using the AWS Management Console. Amazon S3 is an object storage service known for its scalability, data availability, security, and performance (AWS, n.d).

It allows customers to store and protect data for various applications, including websites, mobile apps, backups, and big data analytics. With robust management features and fine-tuned access controls, Amazon S3 meets diverse business and compliance needs and is designed for 99.999999999% (11 9s) durability, supporting millions of applications globally (AWS, n.d).

Lab Scenario

You work for a company that uses Amazon S3 for data storage. An application running on an EC2 instance needs to upload reporting data to an S3 bucket daily. Your task is to create an S3 bucket for your company to store this report data.

To ensure successful deployment, you need to grant the EC2 instance sufficient privileges to upload and retrieve data from the S3 bucket. For security reasons, only the EC2 instance should be allowed to write data to the S3 bucket. Additionally, the files stored in the S3 bucket must be protected against accidental deletion.

Task 1: Create a Bucket

Scenario

You're starting your journey with Amazon S3 to manage EC2 report data, and the first step is to create a new bucket for those reports. Since every object in S3 resides within a bucket, establishing this bucket will allow you to efficiently store and organize your data. Once the bucket is created, you'll have the opportunity to explore and examine various configuration options available for the bucket to enhance its features and security settings.

- ⇒ **Step 1:** Create bucket
- ⇒ **Step 2:** Set "General Configuration" section as reportbucket-AccountID
- ⇒ **Step 3:** Enable ACLS and Object Writer
- ⇒ **Step 4:** Leave the region as its default
- ⇒ **Step 5:** Click "Create bucket"

Create S3 bucket | S3 | us-east-1

us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1&bucketType=general

Account ID: 2604-9755-3246
AWS Labs User: oQN862Zn5rbwVvBHvYJQ9Nd/e7950...

Amazon S3 > Buckets > Create bucket

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

reportbucket-260497553246

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☐ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠️ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

Figure 1: Create bucket page (source: personal collection)

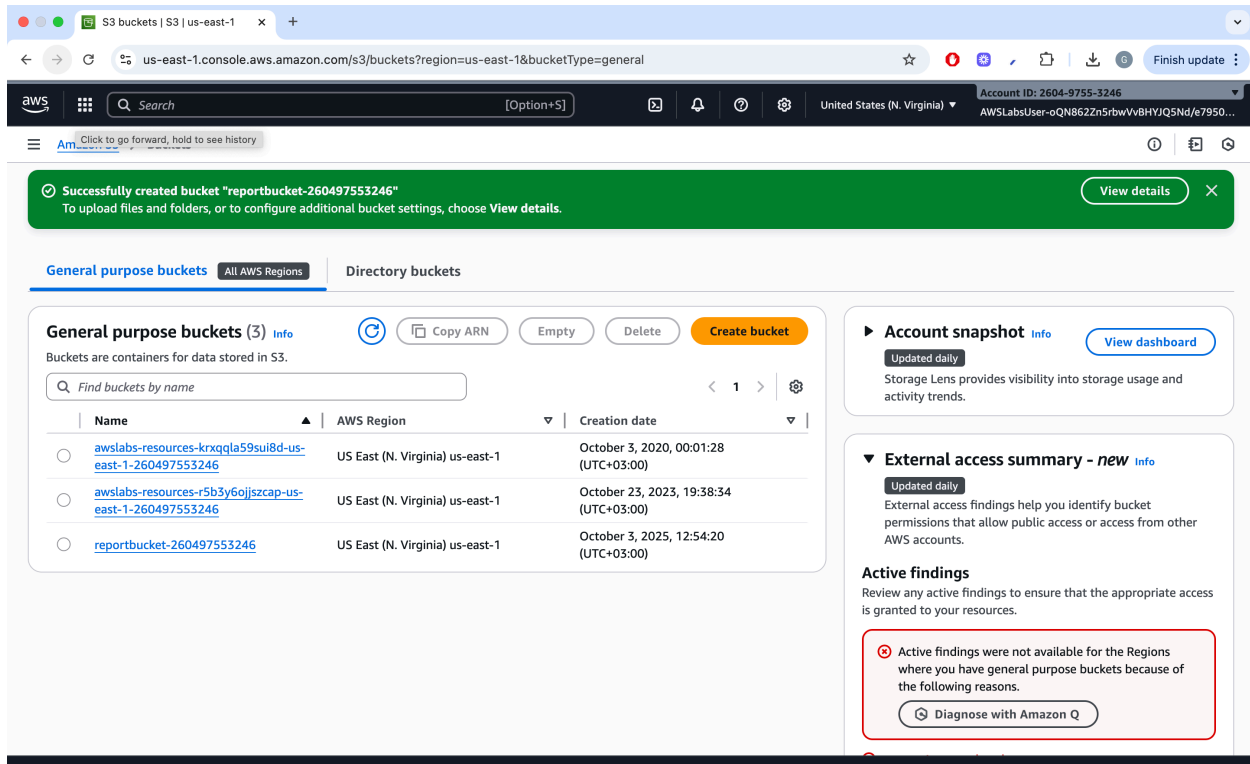


Figure 2: Created S3 Bucket overview page (source: personal collection)

Task 2: Upload an object to the bucket

Scenario

Now that you've set up a bucket for your report data, you're prepared to manage objects. An object can take on many forms, such as a text file, image, video, zip file, etc. When you upload an object to Amazon S3, you have the choice to add metadata along with the object and specify permissions to regulate who can access it. In this task, you'll be testing the upload of objects to your reportbucket. You have a daily report screenshot that you want to upload to your S3 bucket.

- ⇒ **Step 1:** upload a "new-report.png"
- ⇒ **Step 2:** "Close"

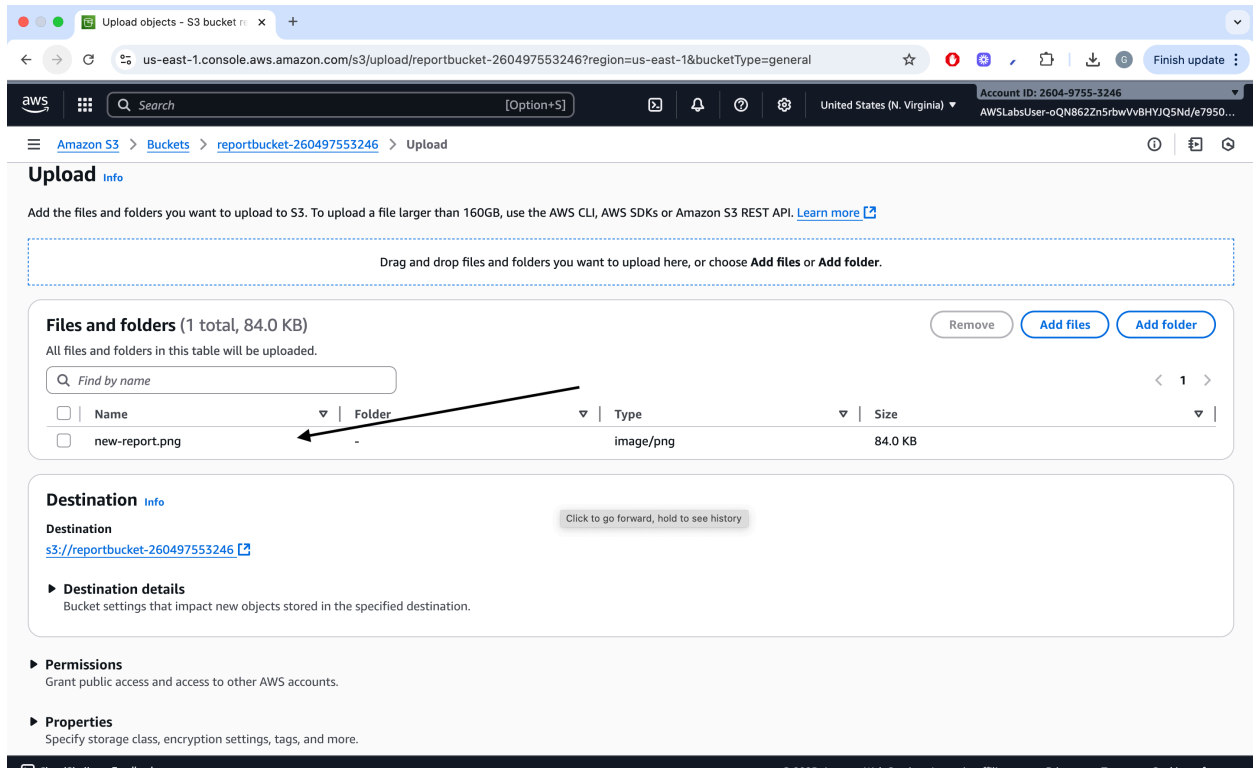


Figure 3: File upload (source: personal collection)

Task 3: Make an object public

Security is prioritized in Amazon S3, so before configuring the EC2 instance to connect to the reportbucket, we connect the bucket and object settings for security. Amazon AWS keeps access to objects private by default.

- ⇒ **Step 1:** Open the reportbucket overview page, and select “new-report.png”
- ⇒ **Step 2:** Locate and copy the “Object URL” link
- ⇒ **Step 3:** Open the link into a new tab
- ⇒ **Step 4:** Notice, the access is denied

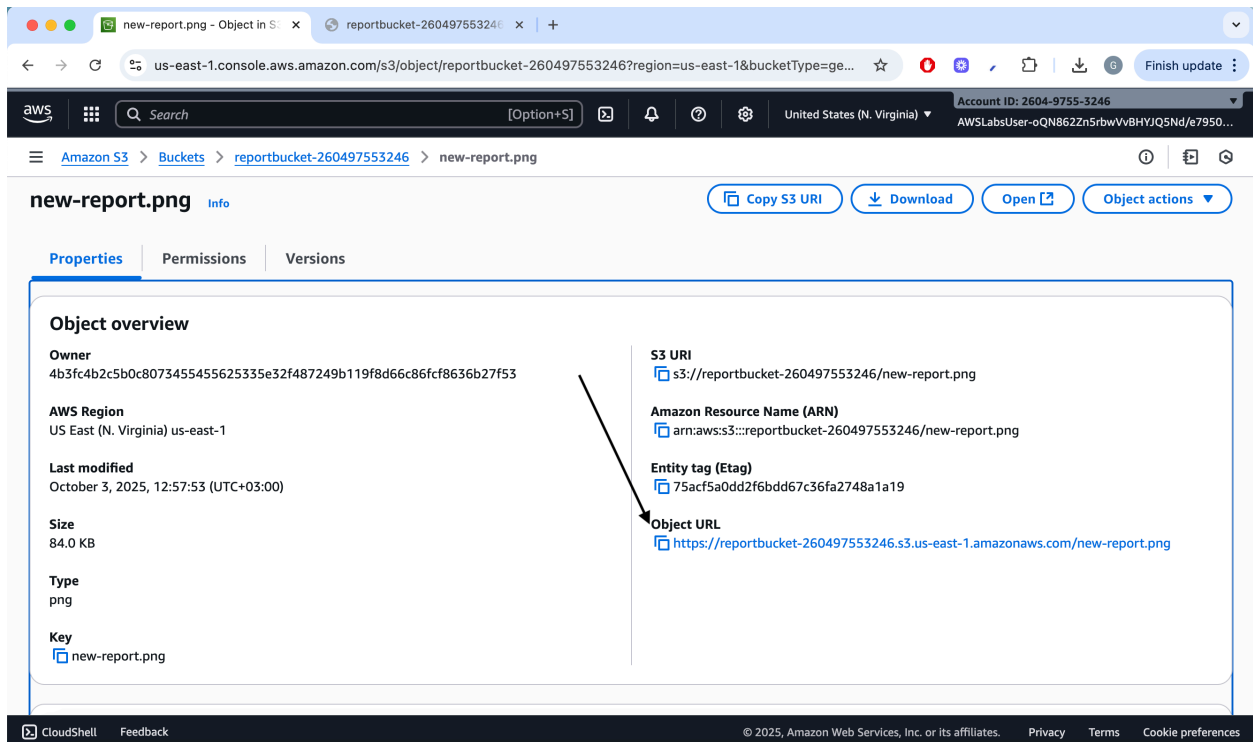


Figure 4: new-report.png object URL (source: personal collection)

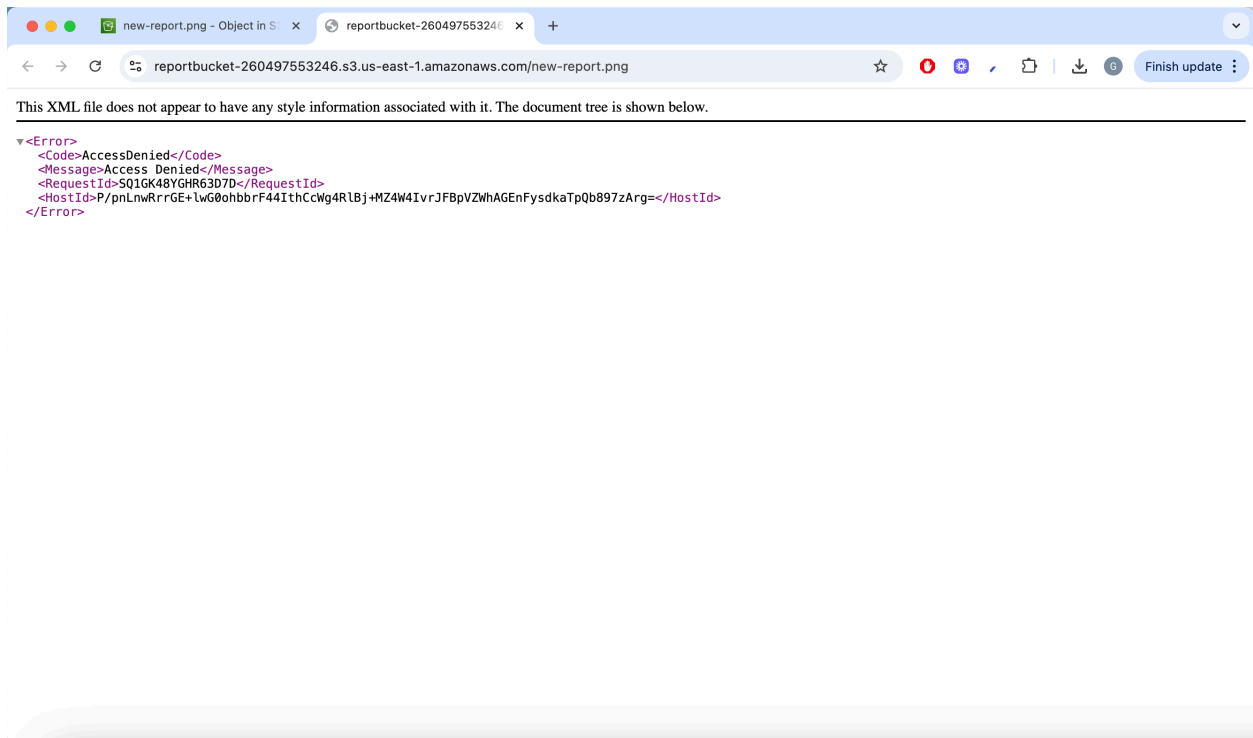


Figure 5: Access denied (source: personal collection)

- ⇒ **Step 5:** Keep the browser tab opened, and make the object public using ACL
- ⇒ **Step 6:** Notice, the action “Failed to edit public access” due to the BPA being enabled.
- ⇒ **Step 7:** Choose the “Permissions” tab

- ⇒ **Step 8:** Under “Block public access (bucket settings), and choose “edit” to change settings
- ⇒ **Step 9:** Deselect the “Block all public access” option, and leave all other options deselected.
- ⇒ **Step 10:** Save changes
- ⇒ **Step 11:** Type “confirm” in the dialogue box

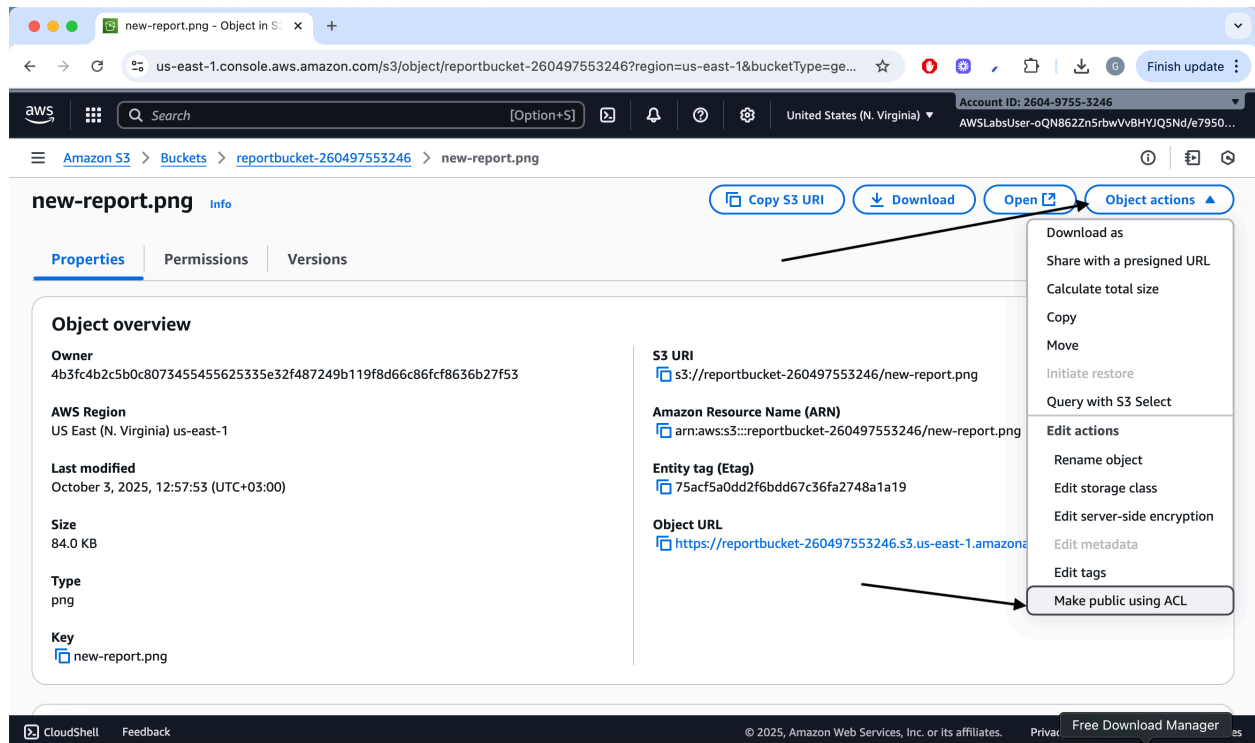


Figure 6: Making the object public (source: personal collection)

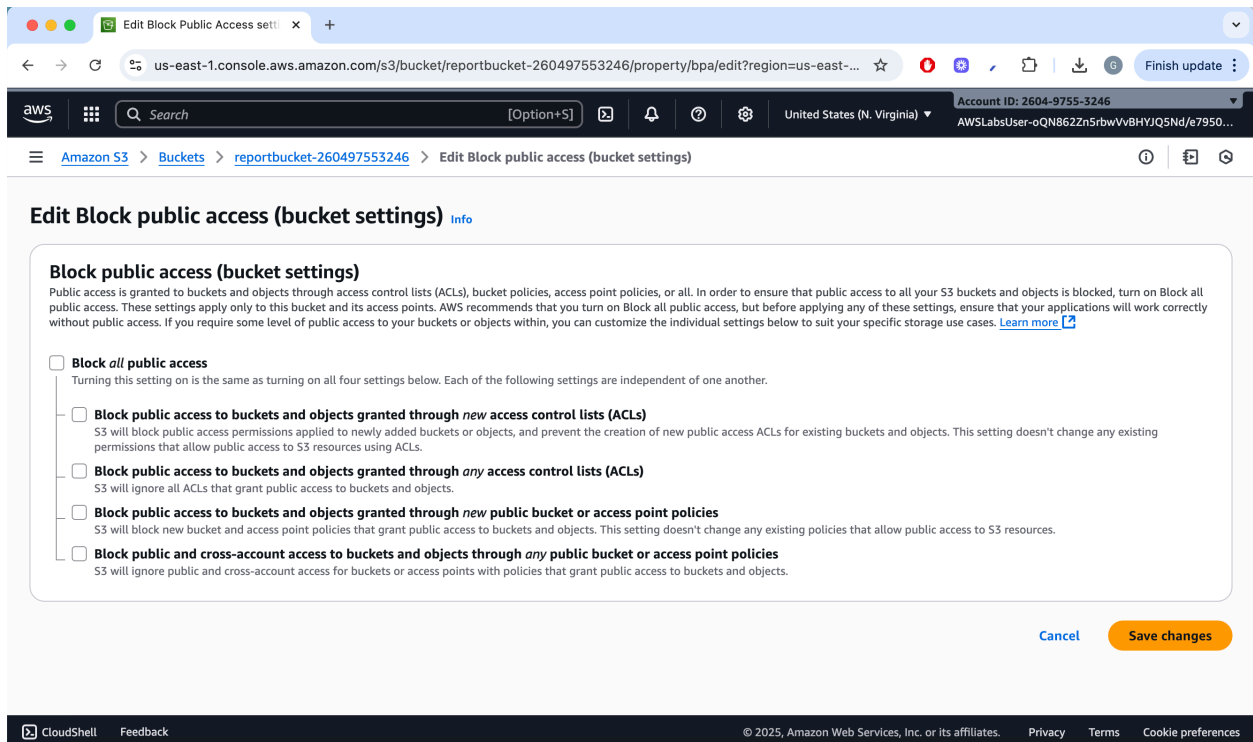


Figure 7: Editing the Block public access (source: personal collection)

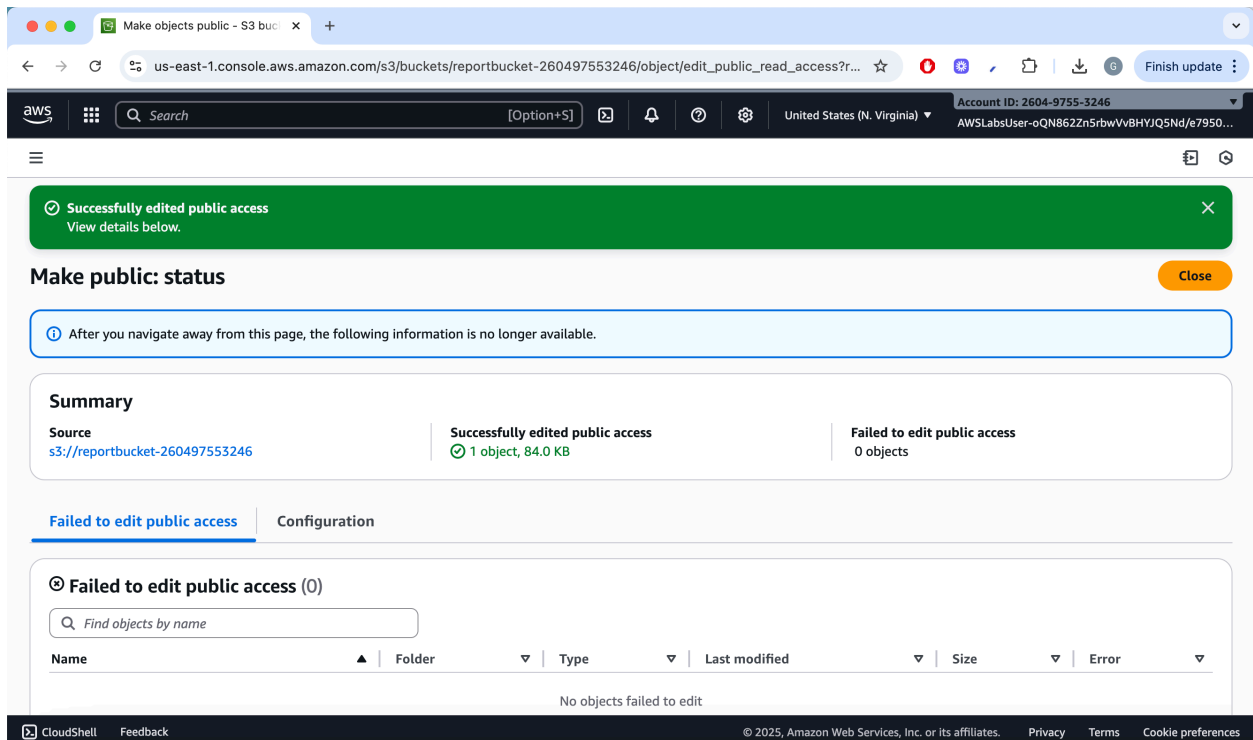


Figure 8: Make object public (source: personal collection)

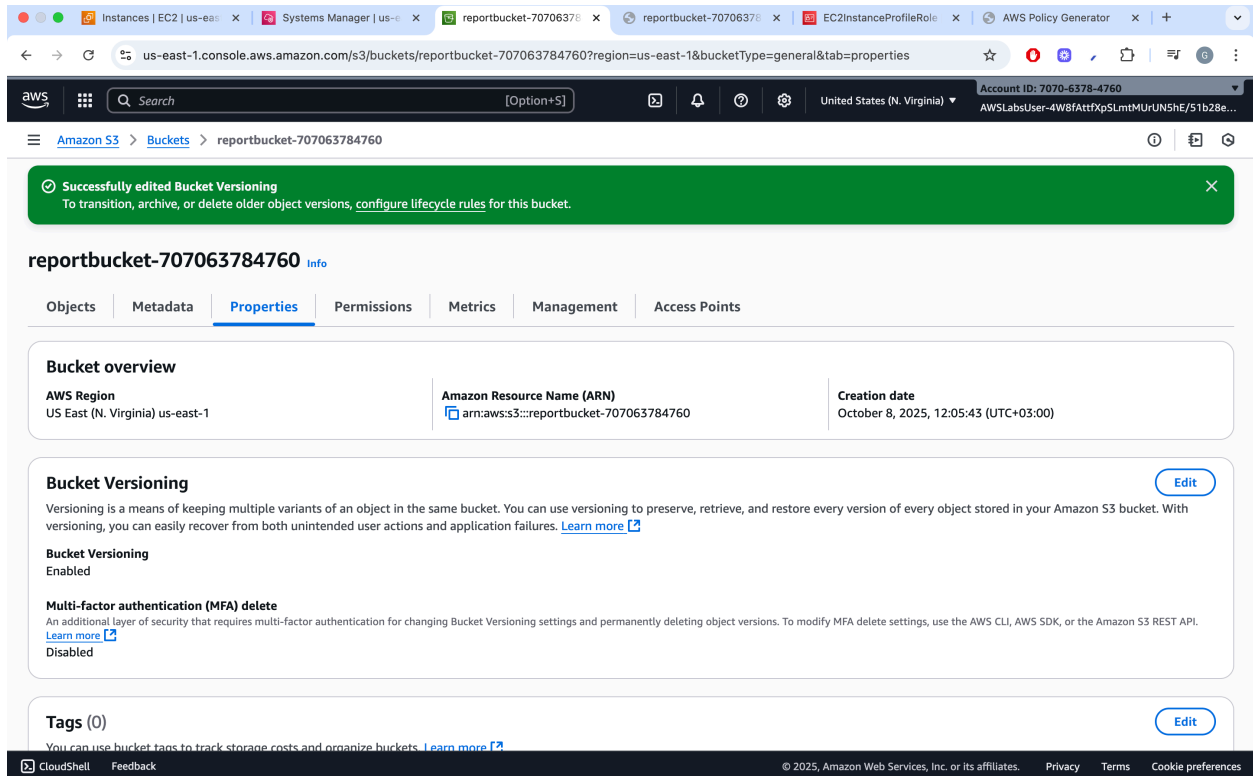


Figure 9: Edited bucket successfully (source: personal collection)

- ⇒ **Step 12:** Reopen the object's tab in the browser
- ⇒ **Step 13:** Close the tab

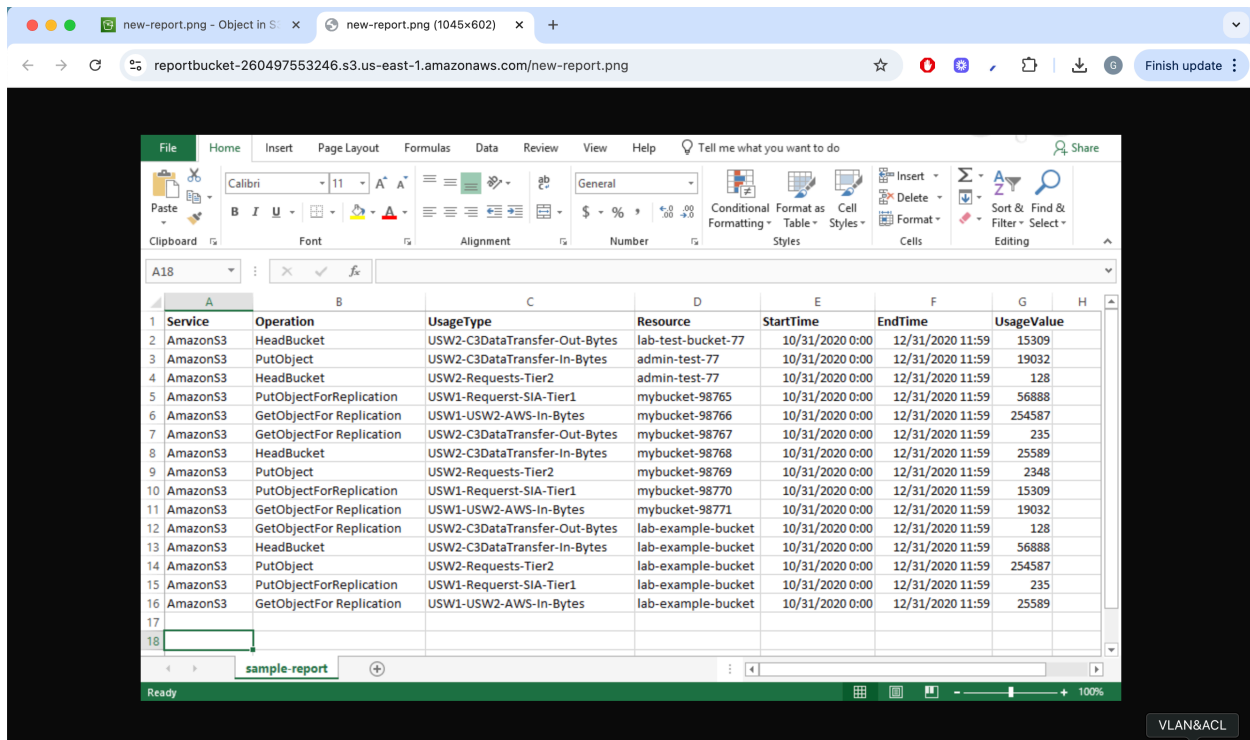


Figure 10: Object file accessible (source: personal collection)

Task 4: Test connectivity from the EC2 instance

In this task, we connected the **Amazon Elastic Compute Cloud (EC2)** instance to test connectivity and security to the S3 reportbucket.

- ⇒ **Step 1:** Under the “Resources” section, choose “Instances (running)”
- ⇒ **Step 2:** Select “Bastion Host” and choose “Connect”
- ⇒ **Step 3:** In the “Connect to instance” window, select “Session Manager”
- ⇒ **Step 4:** Choose “Connect”

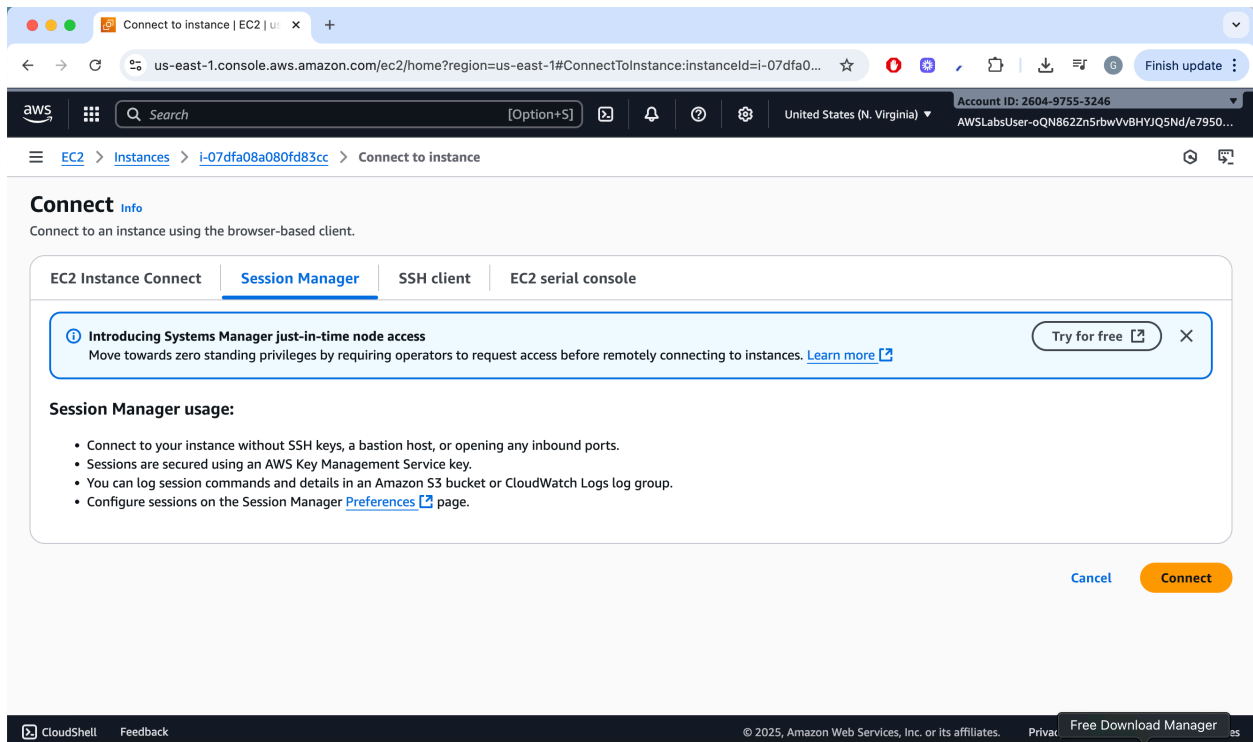
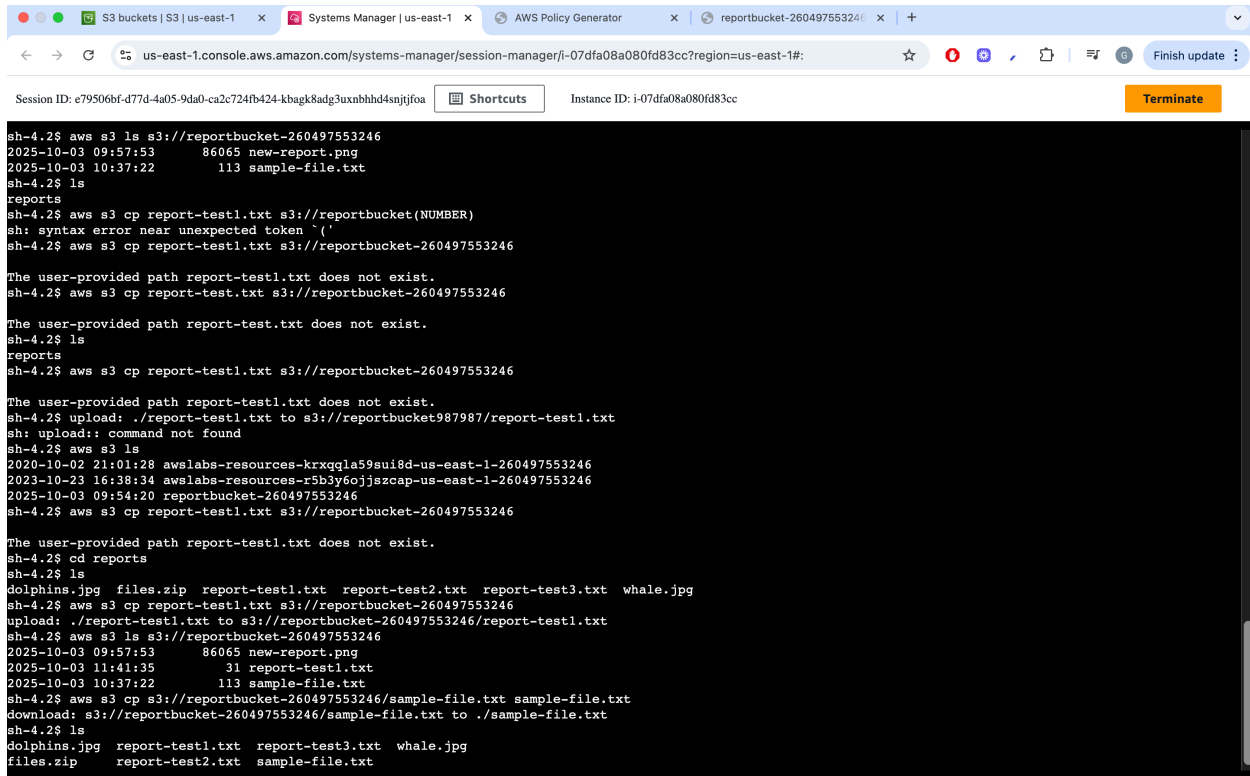


Figure 11: Connect tab overview (source: personal collection)

- ⇒ Step 5: In the bastion host session, enter the command `"cd ~"` to change to home directory (/home/ssm-user/)
- ⇒ Step 6: Verify if you are in the home directory with the command: `"pwd"`
- ⇒ Step 7: Verify the list of all the S3 buckets available with the command `"aws s3 ls"`
- ⇒ Step 8: Verify the objects of the reportbucket with the command `"aws s3 ls s3://reportbucket(ID-Number)"`
- ⇒ Step 9: Verify the list of contents inside the directory with command `"ls"`
- ⇒ Step 10: Copy file to the S3 bucket with command `"aws s3 cp report-test1.txt s3://reportbucket(ID-Number)"`
- ⇒ Step 11: Expect the upload to fail due read-only rights in the permissions



```
sh-4.2$ aws s3 ls s3://reportbucket-260497553246
2025-10-03 09:57:53      86065 new-report.png
2025-10-03 10:37:22       113 sample-file.txt
sh-4.2$ ls
reports
sh-4.2$ aws s3 cp report-test1.txt s3://reportbucket(NUMBER)
sh: syntax error near unexpected token `('
sh-4.2$ aws s3 cp report-test1.txt s3://reportbucket-260497553246

The user-provided path report-test1.txt does not exist.
sh-4.2$ aws s3 cp report-test.txt s3://reportbucket-260497553246

The user-provided path report-test.txt does not exist.
sh-4.2$ ls
reports
sh-4.2$ aws s3 cp report-test1.txt s3://reportbucket-260497553246

The user-provided path report-test1.txt does not exist.
sh-4.2$ upload: ./report-test1.txt to s3://reportbucket987987/report-test1.txt
sh: upload:: command not found
sh-4.2$ aws s3 ls
2020-10-02 21:01:28 awslabs-resources-krxqqla59sui8d-us-east-1-260497553246
2023-10-23 16:38:34 awslabs-resources-r5b3y6ojjszcap-us-east-1-260497553246
2025-10-03 09:54:20 reportbucket-260497553246
sh-4.2$ aws s3 cp report-test1.txt s3://reportbucket-260497553246

The user-provided path report-test1.txt does not exist.
sh-4.2$ cd reports
sh-4.2$ ls
dolphins.jpg files.zip report-test1.txt report-test2.txt report-test3.txt whale.jpg
sh-4.2$ aws s3 cp report-test1.txt s3://reportbucket-260497553246
upload: ./report-test1.txt to s3://reportbucket-260497553246/report-test1.txt
sh-4.2$ aws s3 ls s3://reportbucket-260497553246
2025-10-03 09:57:53      86065 new-report.png
2025-10-03 11:41:35        31 report-test1.txt
2025-10-03 10:37:22       113 sample-file.txt
sh-4.2$ aws s3 cp s3://reportbucket-260497553246/sample-file.txt sample-file.txt
download: s3://reportbucket-260497553246/sample-file.txt to ./sample-file.txt
sh-4.2$ ls
dolphins.jpg report-test1.txt report-test3.txt whale.jpg
files.zip      report-test2.txt sample-file.txt
```

Figure 12: Session manager console (source: personal collection)

Task 5: Create a bucket policy

A bucket policy is a set of permissions associated with an S3 bucket (AWS, n.d). It controls access to an entire bucket or specific directories within a bucket.

- ⇒ **Step 1:** Upload the file “sample-file.txt”
- ⇒ **Step 2:** Locate the object’s link from the overview page, and copy the “Object URL” link
- ⇒ **Step 3:** Notice the “Access Denied” message
- ⇒ **Step 4:** Keep the browser tab opened
- ⇒ **Step 5:** Access the “IAM” then “Roles”
- ⇒ **Step 6:** Copy the EC2 instance’s “Role ARN”
- ⇒ **Step 7:** Choose the “Permissions” tab

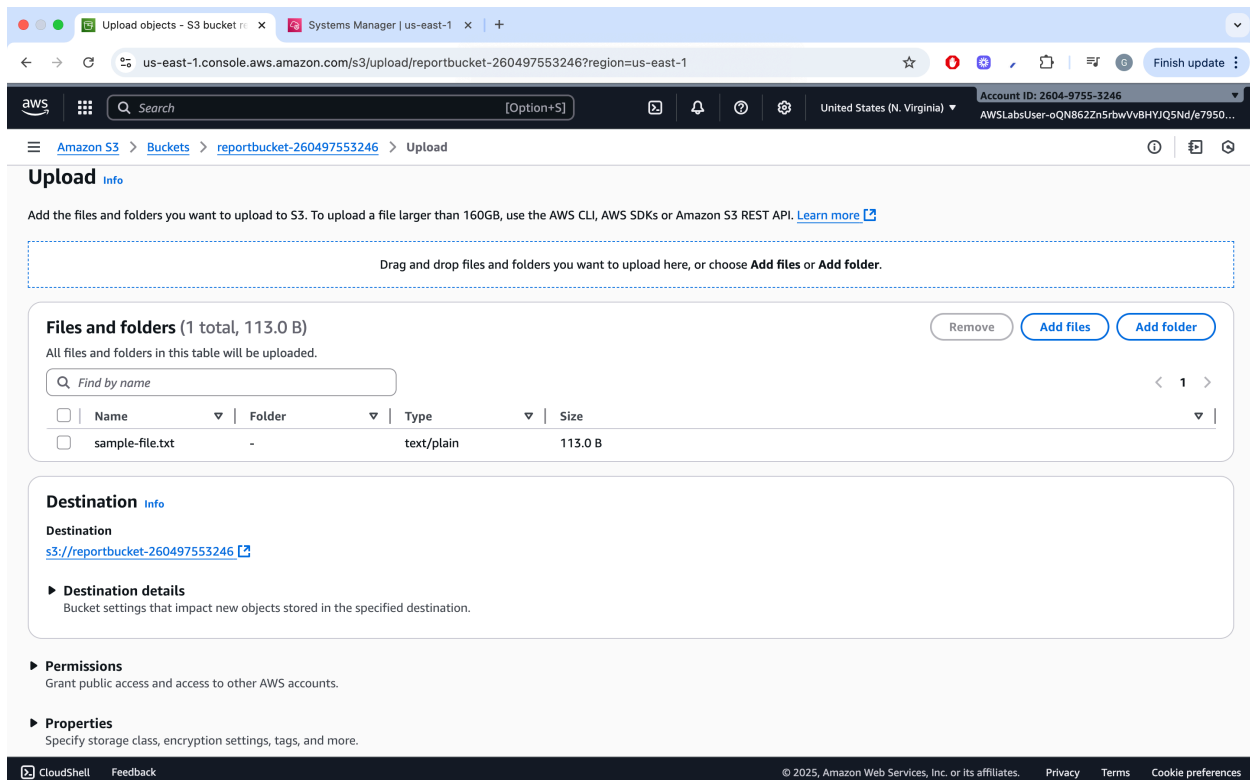


Figure 13: Sample file uploaded (source: personal collection)

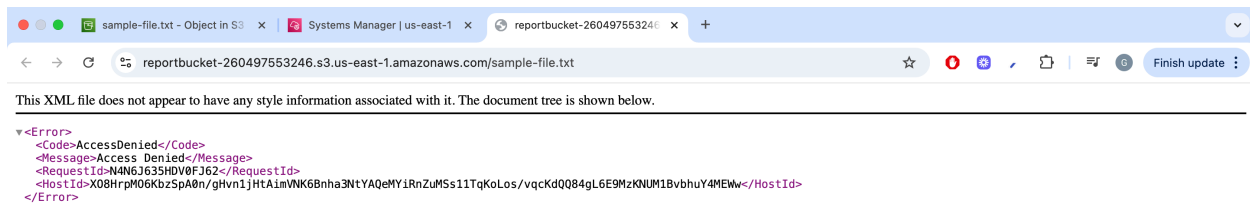


Figure 14: Access denied (source: personal collection)

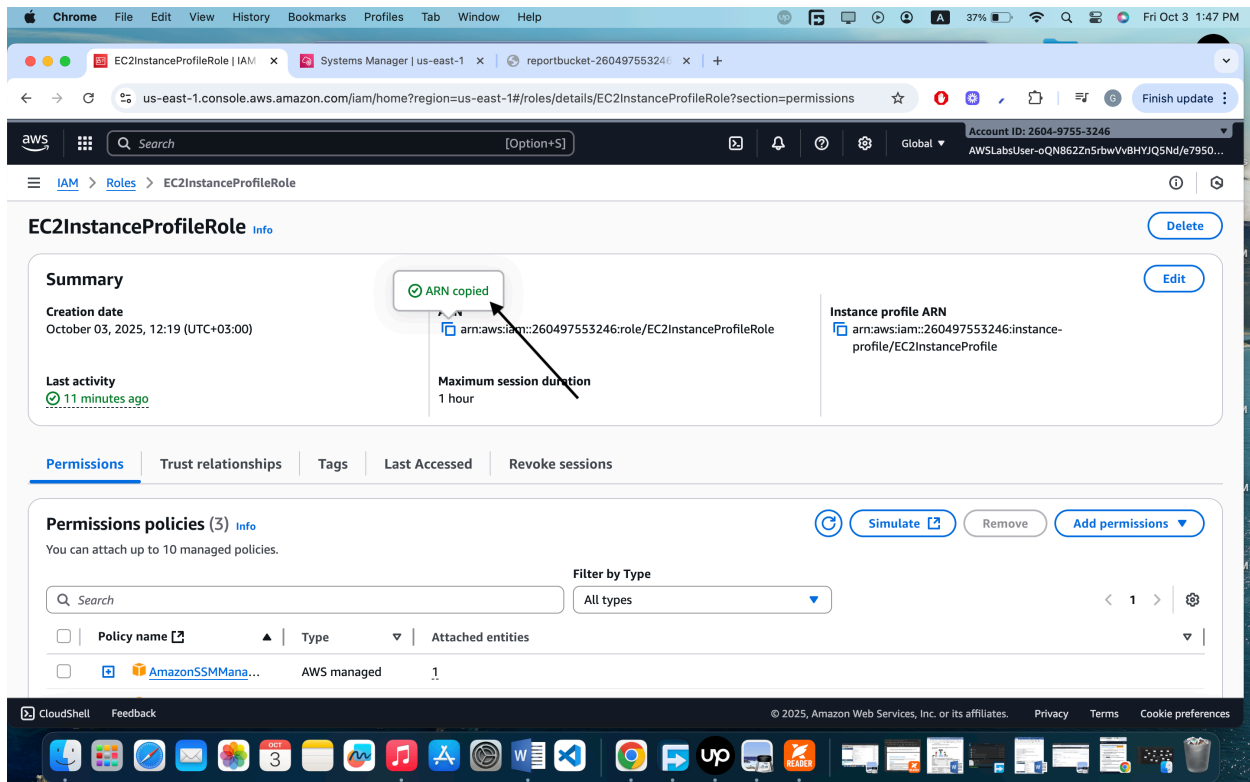


Figure 15: EC2 instance ARN detail (source: personal collection)

- ⇒ **Step 8:** Click on “Policy Generator”
- ⇒ **Step 9:** Select “S3 Bucket Policy”
- ⇒ **Step 10:** Select “Allow” in “Effect”
- ⇒ **Step 11:** For “Principal” use the ARN copied earlier
- ⇒ **Step 12:** Keep default settings for “AWS Service”
- ⇒ **Step 13:** For “Actions”, select “PutObject” and “GetObject”
- ⇒ **Step 14:** Choose “Add statement”
- ⇒ **Step 15:** Choose “Generate Policy”
- ⇒ **Step 16:** Copy the policy and paste it into the “Bucket policy editor”
- ⇒ **Step 17:** Click “Save Changes”

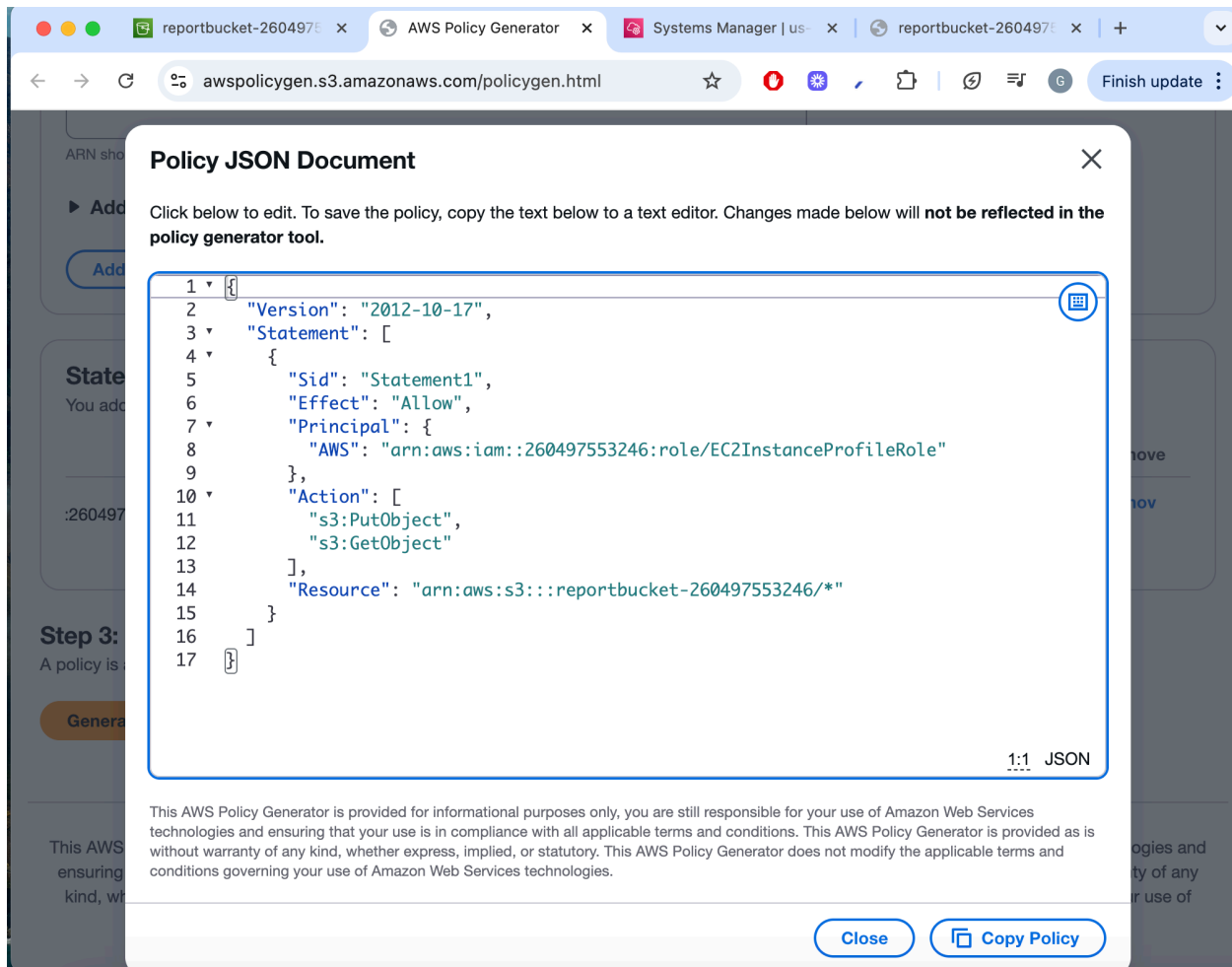
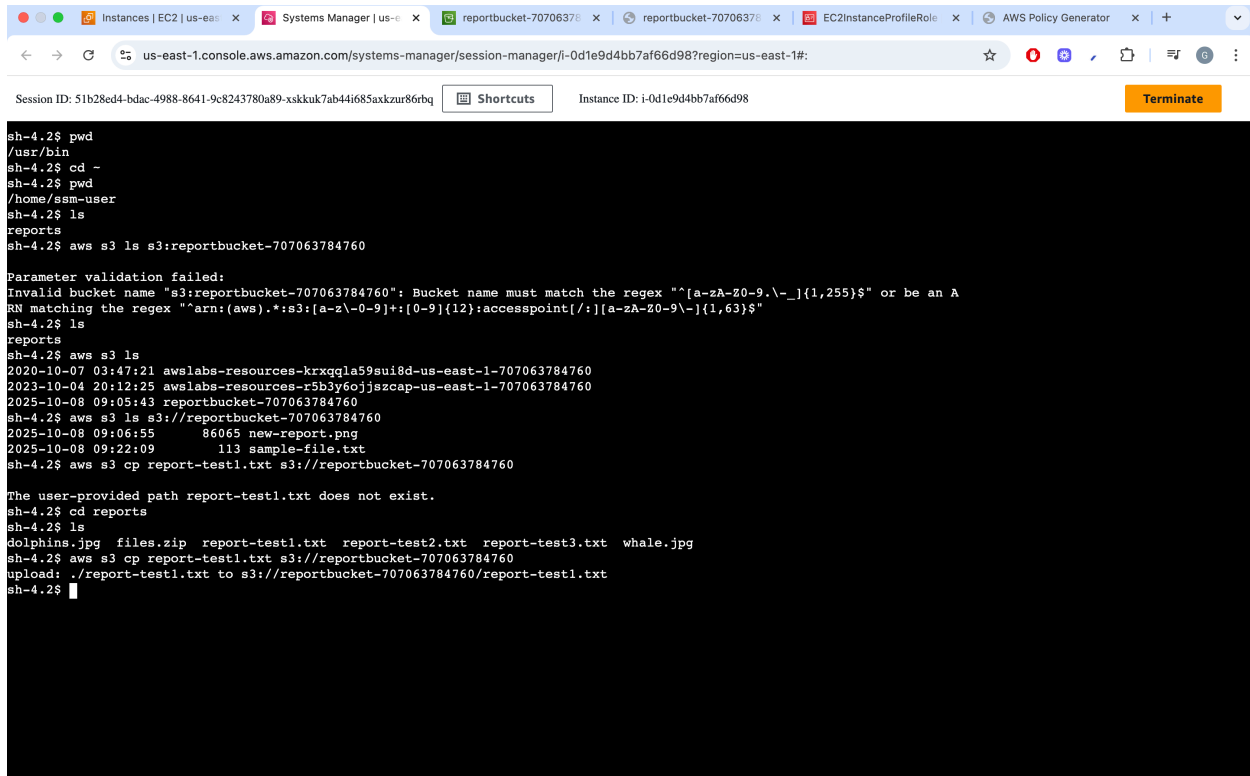


Figure 16: Policy JSON document (source: personal collection)

- ⇒ **Step 18:** From the AWS Systems Manager (SSM), type the command `"aws s3 cp report-test1.txt s3://reportbucket (ID-Number)"`
- ⇒ **Step 19:** Verify if the file is uploaded successfully to S3 with the command `"aws s3 ls s3://reportbucket (ID-Number)"`
- ⇒ **Step 20:** Retrieve a file from the S3 to the EC2 Instance with the command `"aws s3 cp s3://reportbucket (ID-Number) /sample-file.txt sample-file.txt"`
- ⇒ **Step 21:** Verify if the file is available in the reports' directory with the command `"ls"`
- ⇒ **Step 22:** Add a new statement to the policy generator to allow everyone (*), read access (GetObject) .

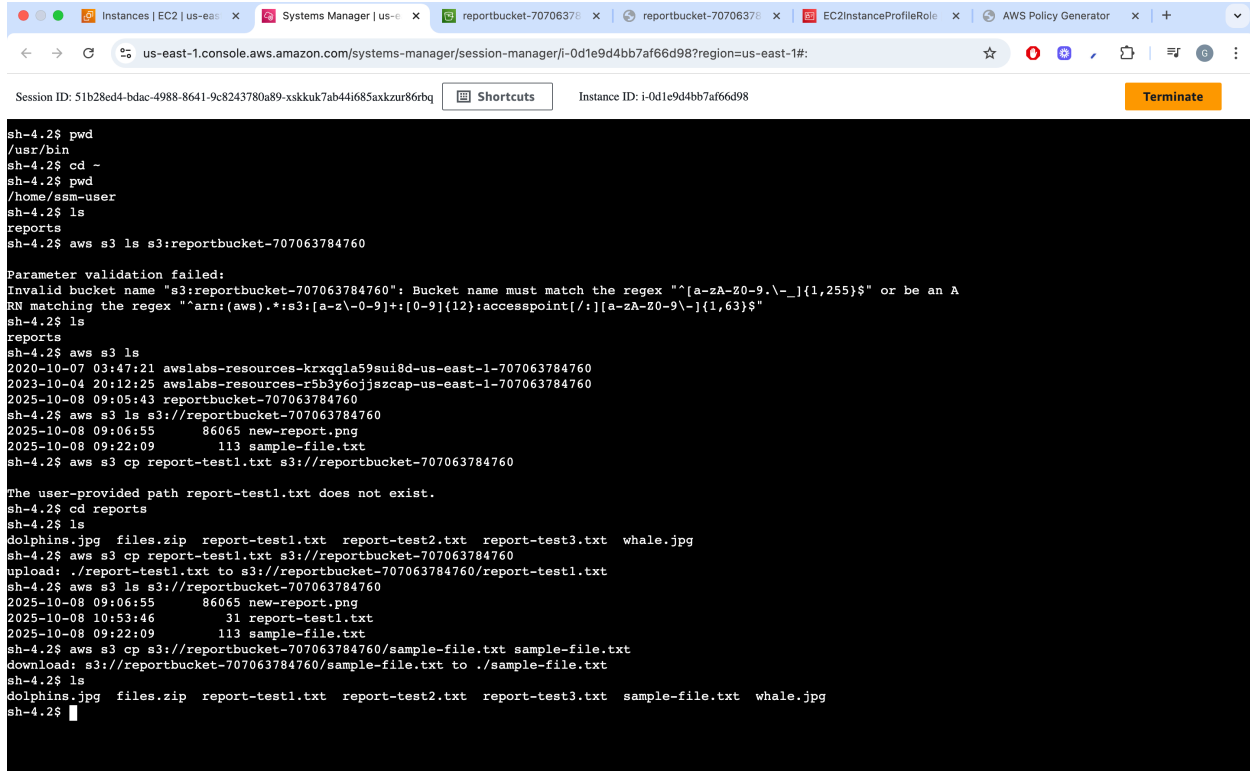


```
sh-4.2$ pwd
/usr/bin
sh-4.2$ cd -
sh-4.2$ pwd
/home/ssm-user
sh-4.2$ ls
reports
sh-4.2$ aws s3 ls s3:reportbucket-707063784760

Parameter validation failed:
Invalid bucket name "s3:reportbucket-707063784760": Bucket name must match the regex "[a-zA-Z0-9.\-]{1,255}" or be an ARN matching the regex "arn:(aws).*:s3:[a-z-0-9]+:([0-9]{12}):accesspoint[/:][a-zA-Z0-9-]{1,63}"
sh-4.2$ ls
reports
sh-4.2$ aws s3 ls
2020-10-07 03:47:21 awslabs-resources-krxqqla59sui8d-us-east-1-707063784760
2023-10-04 20:12:25 awslabs-resources-r5b3y6ojjszcap-us-east-1-707063784760
2025-10-08 09:05:43 reportbucket-707063784760
sh-4.2$ aws s3 ls s3://reportbucket-707063784760
2025-10-08 09:06:55      86065 new-report.png
2025-10-08 09:22:09      113 sample-file.txt
sh-4.2$ aws s3 cp report-test1.txt s3://reportbucket-707063784760

The user-provided path report-test1.txt does not exist.
sh-4.2$ cd reports
sh-4.2$ ls
dolphins.jpg  files.zip  report-test1.txt  report-test2.txt  report-test3.txt  whale.jpg
sh-4.2$ aws s3 cp report-test1.txt s3://reportbucket-707063784760
upload: ./report-test1.txt to s3://reportbucket-707063784760/report-test1.txt
sh-4.2$
```

Figure 17: Copied file overview (source: personal collection)



```
sh-4.2$ pwd
/usr/bin
sh-4.2$ cd -
sh-4.2$ pwd
/home/ssm-user
sh-4.2$ ls
reports
sh-4.2$ aws s3 ls s3:reportbucket-707063784760

Parameter validation failed:
Invalid bucket name "s3:reportbucket-707063784760": Bucket name must match the regex "[a-zA-Z0-9.\-]{1,255}" or be an ARN matching the regex "arn:(aws).*:s3:[a-z-0-9]+:([0-9]{12}):accesspoint[/:][a-zA-Z0-9-]{1,63}"
sh-4.2$ ls
reports
sh-4.2$ aws s3 ls
2020-10-07 03:47:21 awslabs-resources-krxqqla59sui8d-us-east-1-707063784760
2023-10-04 20:12:25 awslabs-resources-r5b3y6ojjszcap-us-east-1-707063784760
2025-10-08 09:05:43 reportbucket-707063784760
sh-4.2$ aws s3 ls s3://reportbucket-707063784760
2025-10-08 09:06:55      86065 new-report.png
2025-10-08 09:22:09      113 sample-file.txt
sh-4.2$ aws s3 cp report-test1.txt s3://reportbucket-707063784760

The user-provided path report-test1.txt does not exist.
sh-4.2$ cd reports
sh-4.2$ ls
dolphins.jpg  files.zip  report-test1.txt  report-test2.txt  report-test3.txt  whale.jpg
sh-4.2$ aws s3 cp report-test1.txt s3://reportbucket-707063784760
upload: ./report-test1.txt to s3://reportbucket-707063784760/report-test1.txt
sh-4.2$ aws s3 ls s3://reportbucket-707063784760
2025-10-08 09:06:55      86065 new-report.png
2025-10-08 10:53:46       31 report-test1.txt
2025-10-08 09:22:09      113 sample-file.txt
sh-4.2$ aws s3 cp s3://reportbucket-707063784760/sample-file.txt sample-file.txt
download: s3://reportbucket-707063784760/sample-file.txt to ./sample-file.txt
sh-4.2$ ls
dolphins.jpg  files.zip  report-test1.txt  report-test2.txt  report-test3.txt  sample-file.txt  whale.jpg
sh-4.2$
```

Figure 18: Retrieving file from the S3 (source: personal collection)

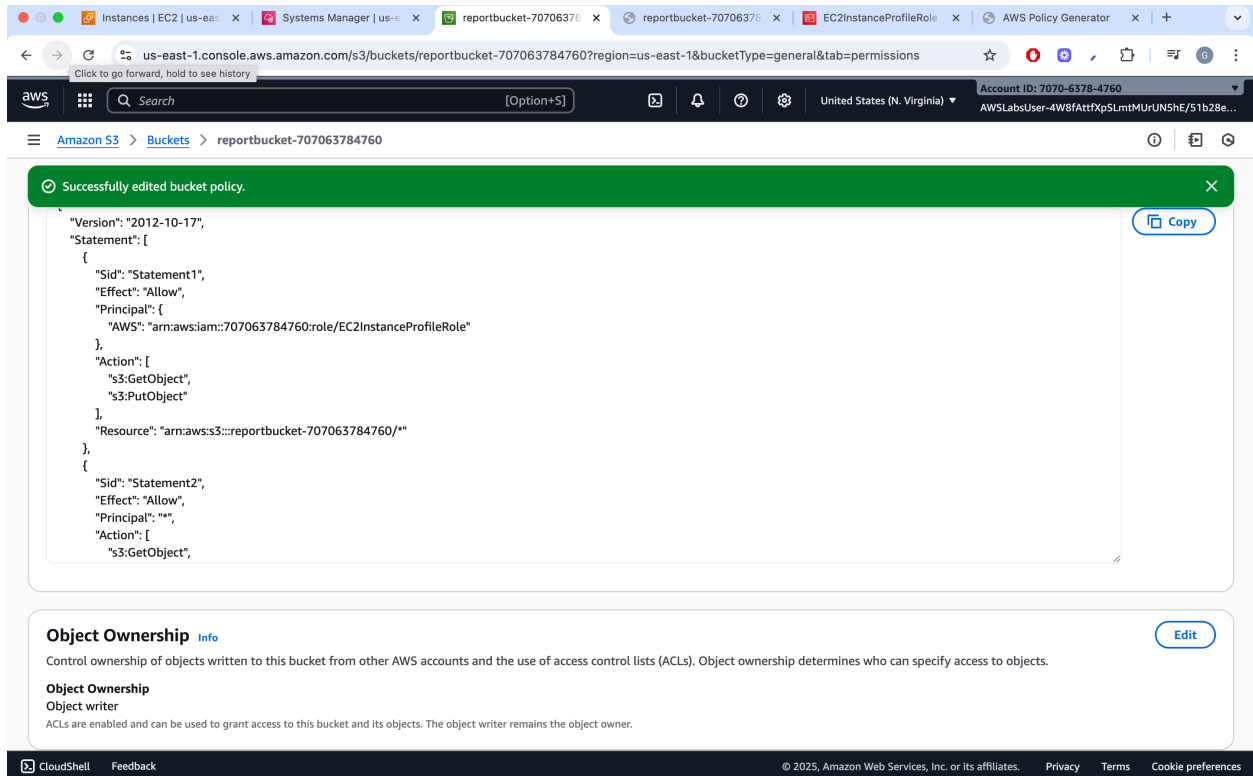


Figure 19: Updated the policy (source: personal collection)

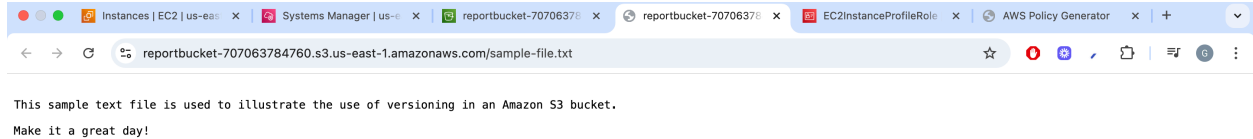


Figure 20: Sample-file.txt display in browser (Source: personal collection)

Task 6: Exploring versioning

This is a way of keeping multiple variants of an object in the same bucket (AWS, n.d). This can be used to preserve, retrieve, and restore every version of every object stored in the Amazon S3 bucket. (AWS, n.d)

- ⇒ **Step 1:** Under the S3 bucket permissions tab, choose “Properties”
- ⇒ **Step 2:** Select “Edit” under the “Bucket Versioning” section
- ⇒ **Step 3:** Enable, and click “Save changes”
- ⇒ **Step 4:** Upload the “Sample-file.txt” file and click “show versions” under the “Objects”

The screenshot shows the AWS Management Console for the bucket 'reportbucket-707063784760'. The 'Objects' tab is selected, displaying a list of four objects. The 'Show versions' toggle is turned on. The objects are:

Name	Type	Version ID	Last modified	Size	Storage class
new-report.png	png	null	October 8, 2025, 12:06:55 (UTC+03:00)	84.0 KB	Standard
report-test1.txt	txt	null	October 8, 2025, 13:53:46 (UTC+03:00)	31.0 B	Standard
sample-file.txt	txt	qED3mTOsgjtRINK2OIWVvXqP35gCGx4G	October 8, 2025, 14:17:11 (UTC+03:00)	171.0 B	Standard
sample-file.txt	txt	null	October 8, 2025, 12:22:09 (UTC+03:00)	113.0 B	Standard

Figure 21: Uploaded a sample file in the s3 bucket (source: personal collection)

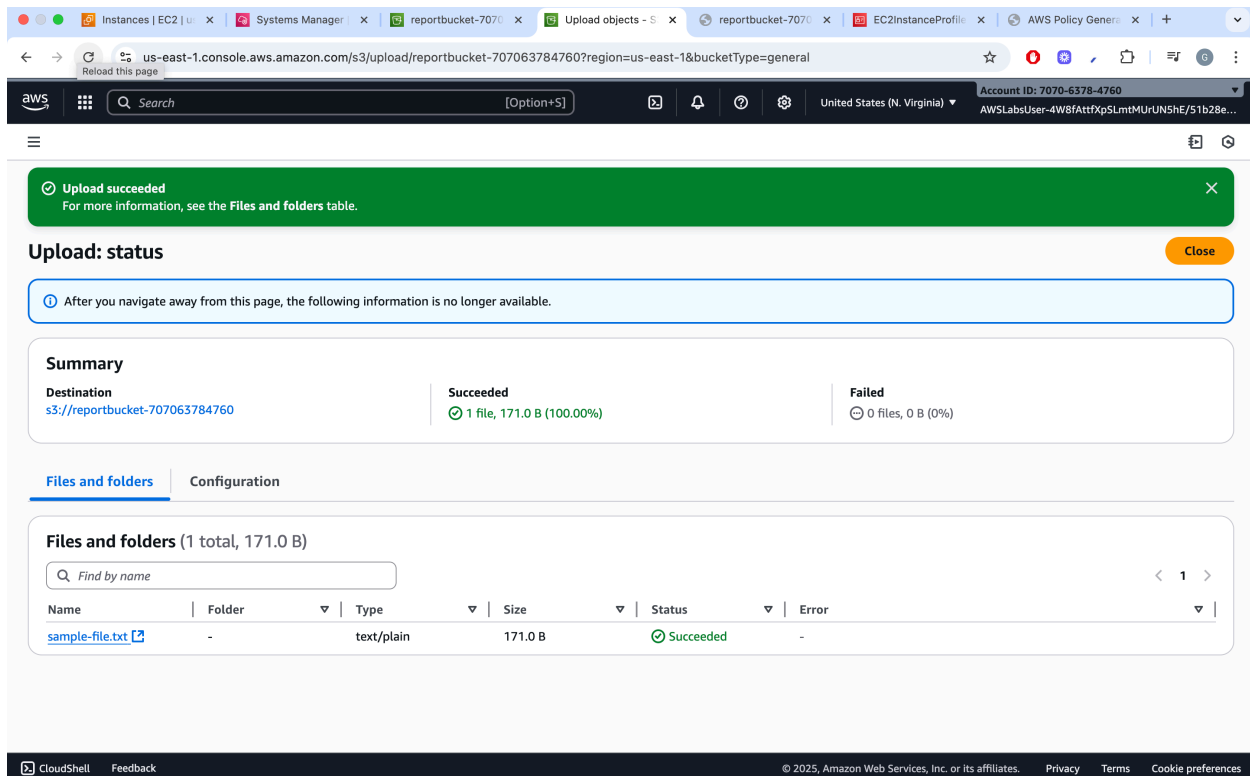


Figure 22: Sample text versioned successfully (Source: personal collection)

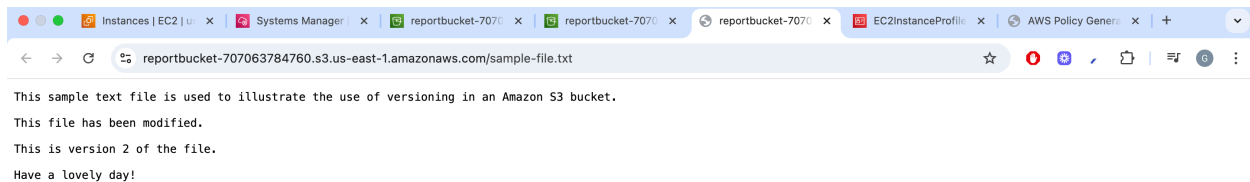


Figure 23: New version displayed (source: personal collection)

Conclusion

In this lab, I created an S3 bucket for a fictional company to use and store report data from the EC2 Instance. I created a bucket policy to allow for the EC2 Instance to PutObject from the reportbucket, and I successfully tested uploading and downloading files from the EC2 instance to test the bucket policy. I then enabled versioning on the S3 bucket to protect against accidental object deletion.

Reference

AWS Training and Certification. (2025, September 25). *Introduction to Amazon Simple Storage Service(S3)*. Retrieved from <https://lab.builder-labs.skillbuilder.aws/sa/lab/arn%3Aaws%3Alearningcontent%3Aus-east-1%3A470679935125%3Ablueprintversion%2FSPL-TF-100-STESS3-2%3A2.4.16-ddb4ea00/en-US>